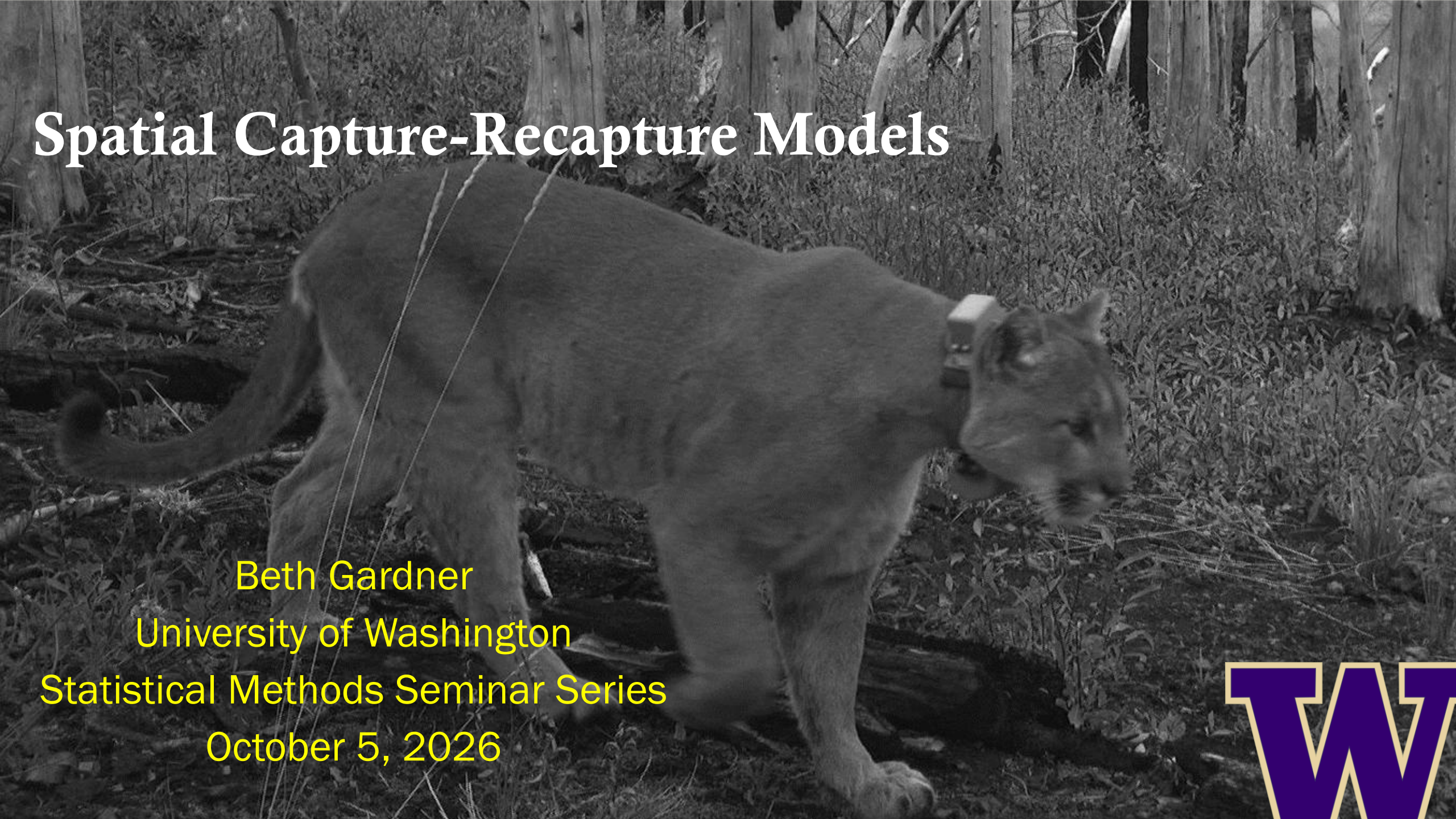


Spatial Capture-Recapture Models



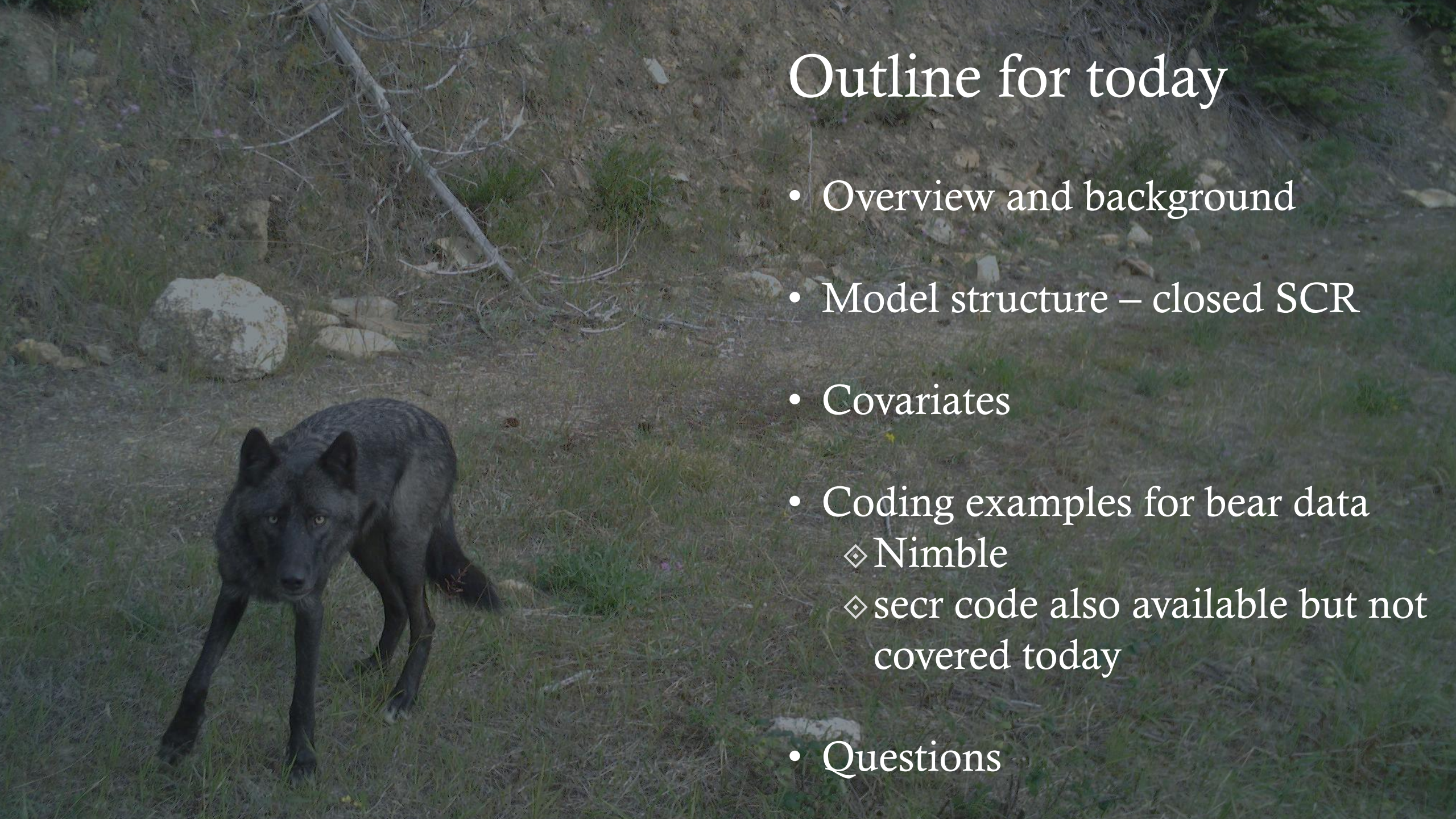
Beth Gardner

University of Washington

Statistical Methods Seminar Series

October 5, 2026



A black wolf is standing in a grassy field, looking towards the camera. In the background, there is a large rock and some dry branches.

Outline for today

- Overview and background
- Model structure – closed SCR
- Covariates
- Coding examples for bear data
 - ◊ Nimble
 - ◊ secr code also available but not covered today
- Questions

SCR overview

- ◆ Demographic questions

- ◆ Abundance (density)

- ◆ Survival, recruitment

- ◆ Ecological questions

- ◆ Space usage

- ◆ Home range size

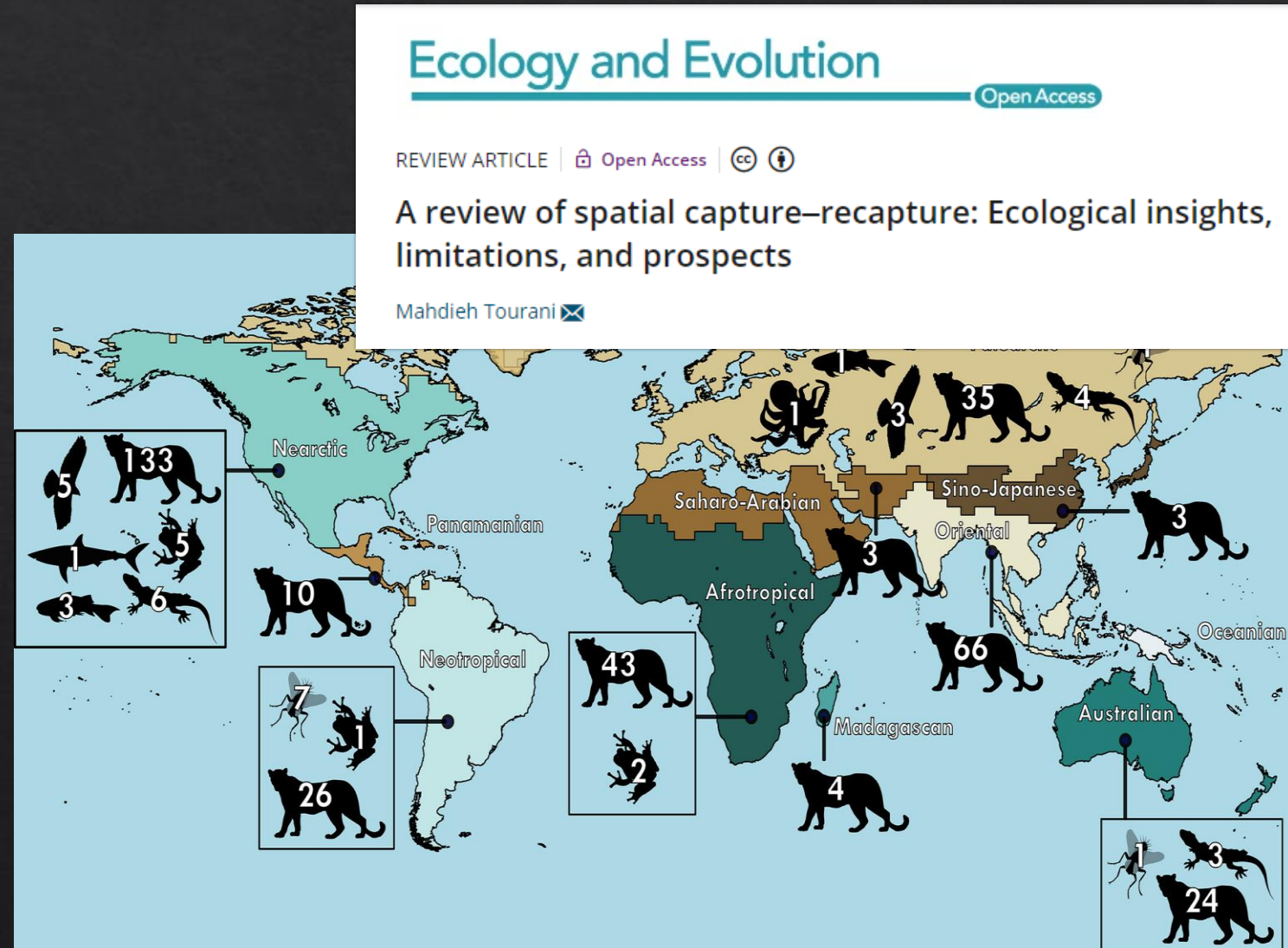
- ◆ Spatial arrangement
of individuals

- ◆ Dispersal, movement



Spatial Capture-Recapture

- ◈ Widely used in ecology to estimate demographic rates in wildlife populations
- ◈ Over 157 species
 - ◈ Invertebrates to whales
 - ◈ 90% focus on mammals



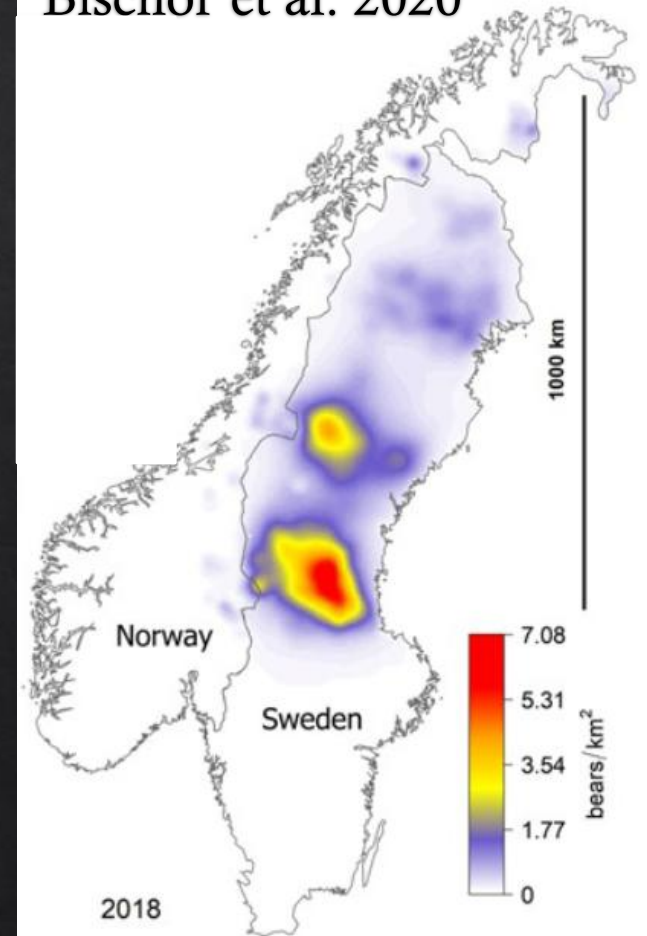
Initial SCR papers

- ◇ 2004 Efford Density estimation in live-trapping studies.
- ◇ 2008 Borchers and Efford Spatially Explicit Maximum Likelihood Methods for Capture-Recapture Studies
- ◇ 2008 Royle and Young A hierarchical model for spatial capture recapture data
- ◇ 2009 Gardner et al. Hierarchical models for estimating density from DNA mark-recapture studies
- ◇ 2009 Royle et al. A hierarchical model for estimating density in camera-trap studies

Spatial capture recapture developments

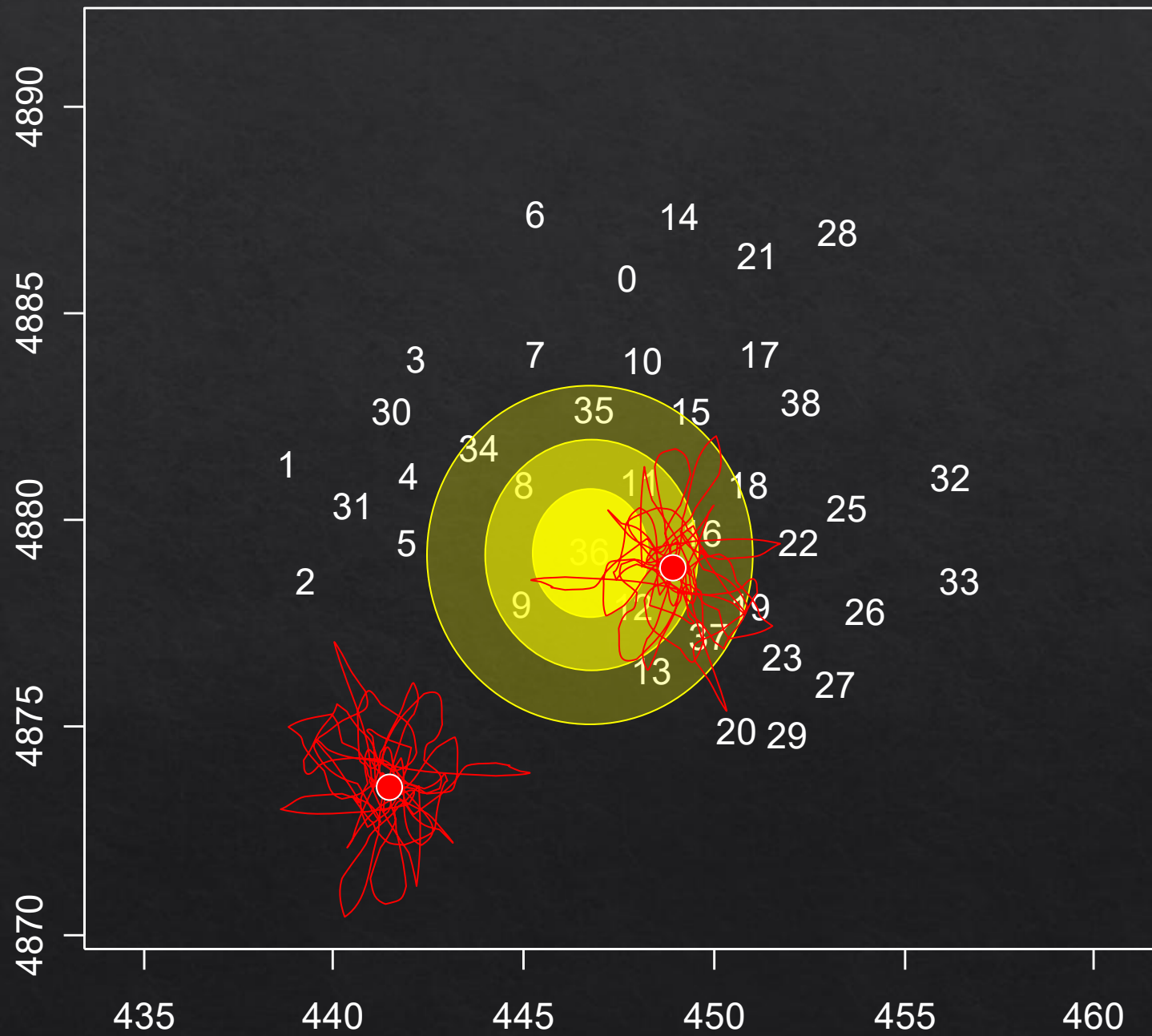
- ◇ Open population models
- ◇ Inhomogeneous point processes
- ◇ Partially identifiable and fully unmarked populations
- ◇ Spatial mark resight
- ◇ Integrating other data sources: telemetry data, dead recoveries,
- ◇ Dispersal
- ◇ Territoriality, avoidance, clustering, etc.
- ◇ Least cost paths
- ◇ Resource selection functions
- ◇ Explicit movement models

Bischof et al. 2020



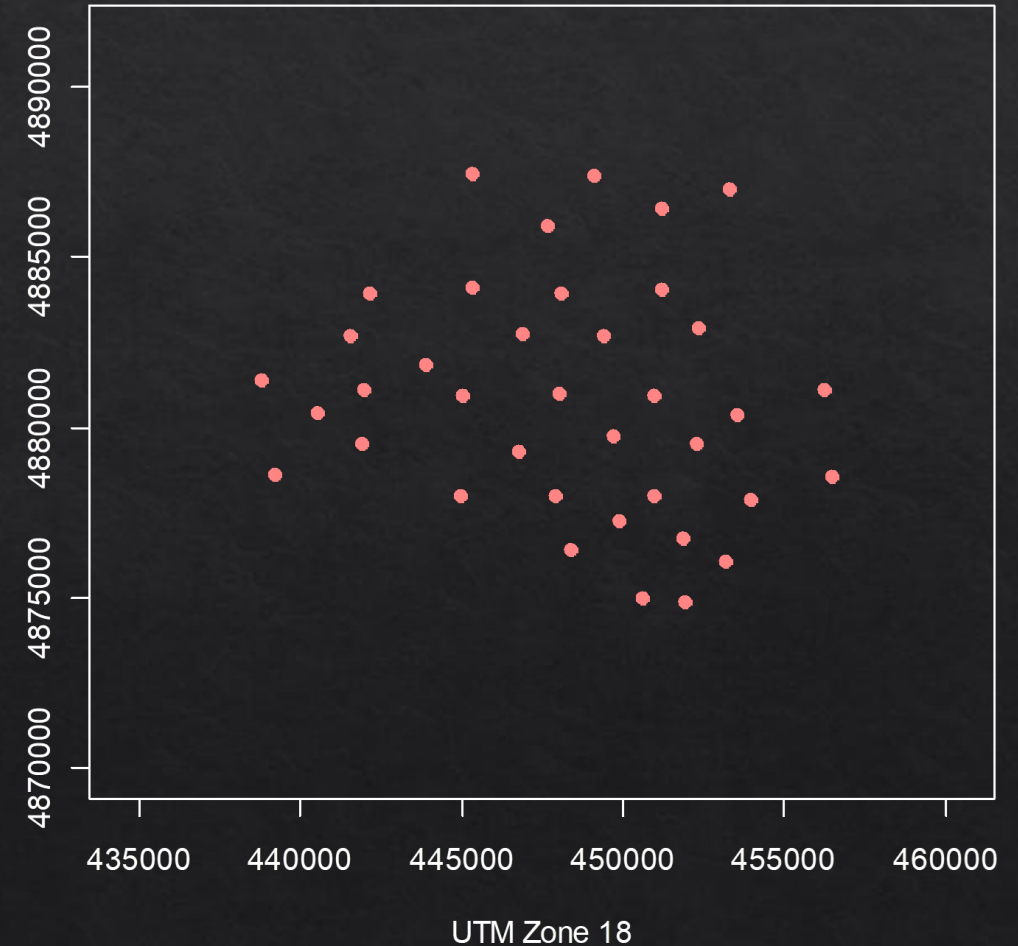
Variable Exposure

- ◆ Individuals have variable exposure to the trap array based on where they “live” (Efford 2004)
 - ◆ Essentially individual heterogeneity governed by a home range or activity center of sorts
- ◆ Leads to heterogeneity in the probability of capture

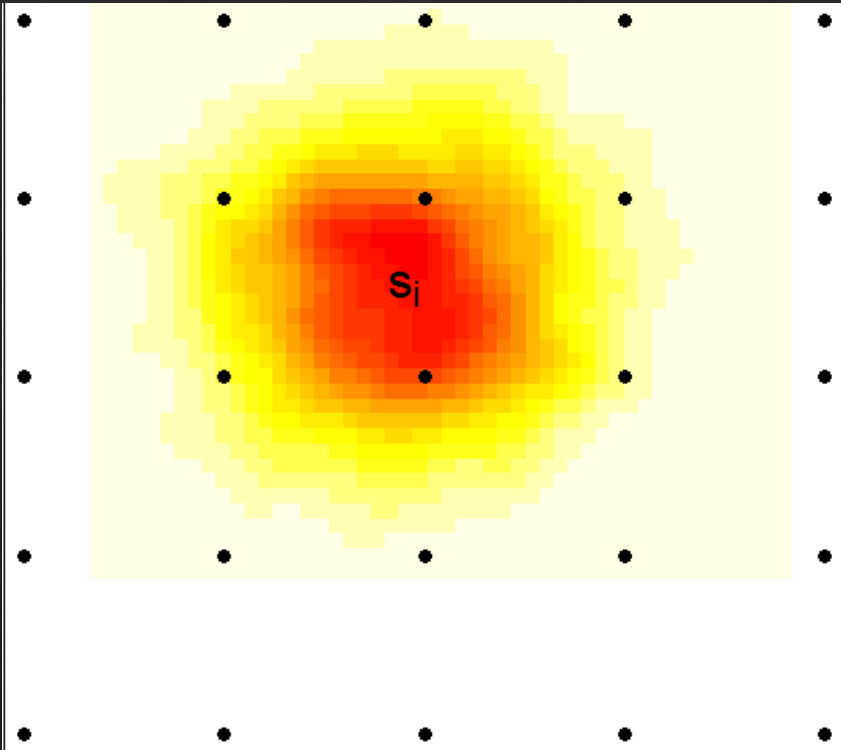


Spatial Capture-Recapture Overview

- ◇ Data are collected via traps placed in an array or transects
 - ◇ Camera traps, hair snares, scat surveys, small mammal traps, etc
- ◇ Build encounter histories (1 = animal i was detected at trap j at time k , 0 otherwise)
- ◇ Use these encounter histories to estimate detection, abundance, demographic rates



Spatial capture-recapture



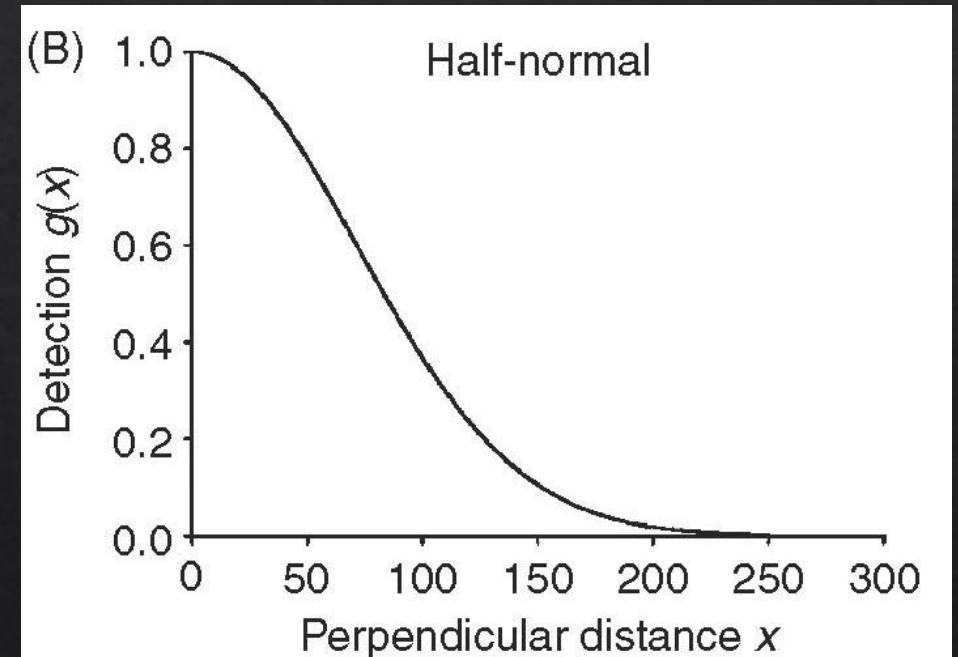
- ◇ Each individual i has an activity center s_i
- ◇ Each trap j has coordinates $\mathbf{x}_j = (x_{1j}, x_{2j})$
- ◇ Encounter rate with trap j ,
$$\lambda_{ij} = f(\text{distance}[\mathbf{x}_j, \mathbf{s}_i])$$
- ◇ f is declining function, governed by detection parameters

Distance Function

- ◆ Define d_{ij} as the distance between trap j ($\mathbf{x}_j$) and individual i 's “activity center” ($\mathbf{s}_i$)

$$d_{ij} = |\mathbf{s}_i - \mathbf{x}_j|$$

- ◆ Half-normal: $g(x) = \exp(-\frac{d_{ij}^2}{2\sigma^2})$



Observation process

- ◆ In standard CMR models, individuals could only be encountered once per sampling occasion
 - ◆ In SCR, animals can potentially be detected at more than one detector during a sampling occasion (e.g., hair snares)
 - ◆ Animals can also potentially be detected multiple times at the same detector or multiple detectors within a sampling occasion (e.g., camera traps)
- ◆ SCR models easily account for different observation process



Sampling methods

- ◇ Examples
 - ◇ Hair snares
 - ◇ Scent lures
 - ◇ Camera traps
 - ◇ Scat surveys
 - ◇ Mist nets
 - ◇ Cover board surveys
 - ◇ PIT tags and antennae
 - ◇ Acoustic arrays

Observation models and detector types

- ◆ **Poisson (detector type termed “count”)**

- ◆ Detections independent between detectors and more than 1 allowed per detector
- ◆ Individuals can be captured multiple times in multiple traps per occasion
- ◆ e.g., camera trapping, scat surveys

- ◆ **Binomial (detector type termed “proximity”)**

- ◆ Detections independent between detectors
- ◆ Individuals can be captured only once in a trap but can be captured in multiple traps per occasion
- ◆ e.g., acoustic surveys, hair snares

- ◆ **Multinomial (detector type termed “multi-catch trap”)**

- ◆ Detectors “compete” for animals
- ◆ Individuals can only be captured in 1 trap per occasion
- ◆ e.g., pitfall traps, mist nests

Observation Model - Poisson

y_{ijk} is the encounter history for individual i in trap j during interval k

$$y_{ijk} \sim \text{Poisson}(\lambda_{ijk})$$

$$\lambda_{ijk} = \lambda_0 \exp\left(\frac{-d_{ij}^2}{2\sigma^2}\right)$$

Observation Model - Binomial

y_{ijk} is the encounter history for individual i in trap j during interval k

$$y_{ijk} \sim \text{Binomial}(g_{ijk}, 1)$$

$$g_{ijk} = g_0 \exp\left(\frac{-d_{ij}^2}{2\sigma^2}\right)$$

Multinomial Model

$$y_{ik} \sim \text{Multinom}(1, \boldsymbol{\pi}(s_i))$$

- ◇ y_{ik} is the trap that individual i was captured in during sample occasion k
- ◇ Where $\boldsymbol{\pi}$ is a vector of length $J+1$, and $\pi_{i,J+1}$, the last cell, corresponds to the probability of the event “not captured”

Multinomial Model

For each trap $j = 1 \dots J$

$$\pi_{ij} = \frac{\exp(g_0 f(d_{ij}))}{1 + \sum_j \exp(g_0 f(d_{ij}))}$$

For “not captured”

$$\pi_{i,j+1} = \frac{1}{1 + \sum_j \exp(g_0 f(d_{ij}))}$$

Covariates

- ◆ Trap level
 - ◆ Baited/not-baited, On/Off Roads, Trap failures, etc.
- ◆ Occasion/survey level
 - ◆ Weather, date, etc.
- ◆ Individual level
 - ◆ Male/Female, Age, Size, recapture, etc.
- ◆ Habitat/Density

Covariates

1. Modify the baseline encounter process
2. Modify the distance function

$$\lambda_{ijk} = \lambda_{0,ijk} \exp\left(\frac{-d_{ij}^2}{2\sigma_i^2}\right)$$

The diagram shows two arrows originating from the $\lambda_{0,ijk}$ term in the equation above. One arrow points down and to the left to the expression $\lambda_0 + \alpha * \text{Cov}_{ijk}$. The other arrow points down and to the right to the expression $\sigma^2[\text{Ind}_i]$.

$$\lambda_0 + \alpha * \text{Cov}_{ijk}$$
$$\sigma^2[\text{Ind}_i]$$

- Individual level
 - Occasion level
 - Trap level
-
- Fixed
 - Latent
 - Observed
 - Partially observed

3. Modify the density function



Model fitting

- ◆ Maximum likelihood (Borchers and Efford, 2008) – integration over all possible activity centers in S
- ◆ Bayesian (Royle et al., 2013) - using the full joint likelihood of detection parameters and activity centers, samples activity centers within an MCMC algorithm
- ◆ Bayesian inference (King et al., 2016) using the semi-complete data likelihood

Data augmentation

- ◆ General extensibility
 - ◆ No additional complications if covariate values were not recorded for some of the observed individuals.
- ◆ Easy to parameterize and fit
- ◆ N is estimated within the model (as opposed to methods based on the conditional likelihood)
 - ◆ Fully model-based formulation of individual covariate models

Data Augmentation

Capture-recapture

Augmented data

j=1	j=2	j=3	j=1	j=2	j=3	
0	1	0	0	1	0	
1	0	1	1	0	1	
0	1	0	0	1	0	
1	0	1	1	0	1	
0	1	1	0	1	1	
0	1	1	0	1	1	
1	1	1	1	1	1	
1	1	1	1	1	1	
						n
			0	0	0	
			0	0	0	
			0	0	0	
						N
			0	0	0	
			.	.	.	
			.	.	.	
						M
			0	0	0	

Data augmentation

- ◆ To set this up, augment the encounter histories with a set of $M - n$ all zeros (Royle, Dorazio, & Link 2007)
- ◆ Some of the all zero encounter histories will be part of the population and the others will be extra – essentially a zero-inflated model, where each individual encounter history follows a Bernoulli process with inclusion parameter ψ

$$z_i \sim \text{Bernoulli}(\psi)$$

Data Augmentation

- ◆ Use a uniform prior on N on the integers up to some big number M

$$N \sim \text{Uniform}(0, M)$$

- ◆ In practice, commonly allow each individual to have an inclusion parameter z , which follows a Bernoulli process with probability ψ , such that

$$N = \sum_{i=1}^M z_i$$

NIMBLE

- ◇ <https://r-nimble.org/>
- ◇ Numerical Inference for statistical Models using Bayesian and Likelihood Estimation.
- ◇ A framework for hierarchical statistical models and methods.
- ◇ A configurable system for MCMC.
- ◇ A nearly drop-in alternative to WinBUGS, OpenBUGS and JAGS and an extension for:
 - ◇ writing new functions and distributions.
 - ◇ named alternative parameterizations (e.g. sd vs precision for ``dnorm``).
 - ◇ additional R-like coding, including vectorized declarations.

A library of other methods. –Particle MCMC, RJMCMC, CAR models, etc

This info used from Lauren Ponisio's presentation on NIMBLE

There is also a nimbleSCR package, which I will not cover today, but can be used to speed up your SCR models!



◆ Worked examples

- ◆ All data and scripts are on github

- ◆ Bear dataset

- ◆ Move to R

Black bears on Fort Drum



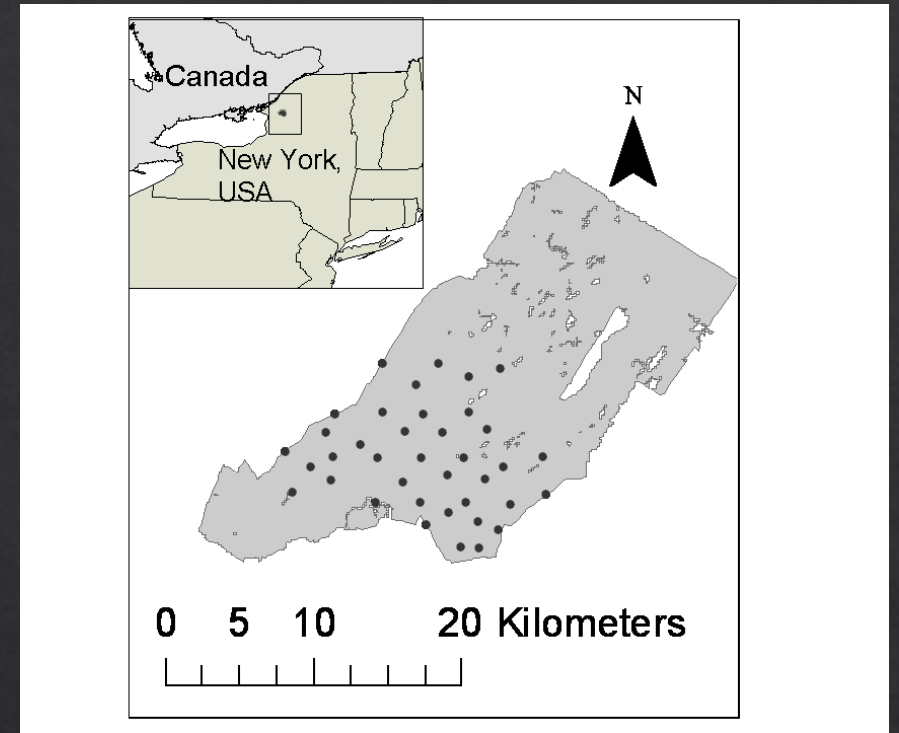
- ◇ DNA based capture-recapture study using hair snares
- ◇ Estimate the density bears on Fort Drum Military Base

Gardner, B., Royle, J.A., Wegan, M.T., Rainbolt, R.E., and Curtis, P.D. 2010. Density estimation based on hair snare data for black bears. *Journal of Wildlife Management*. 74(2):318-325.

<https://doi.org/10.2193/2009-101>

Bear data collected

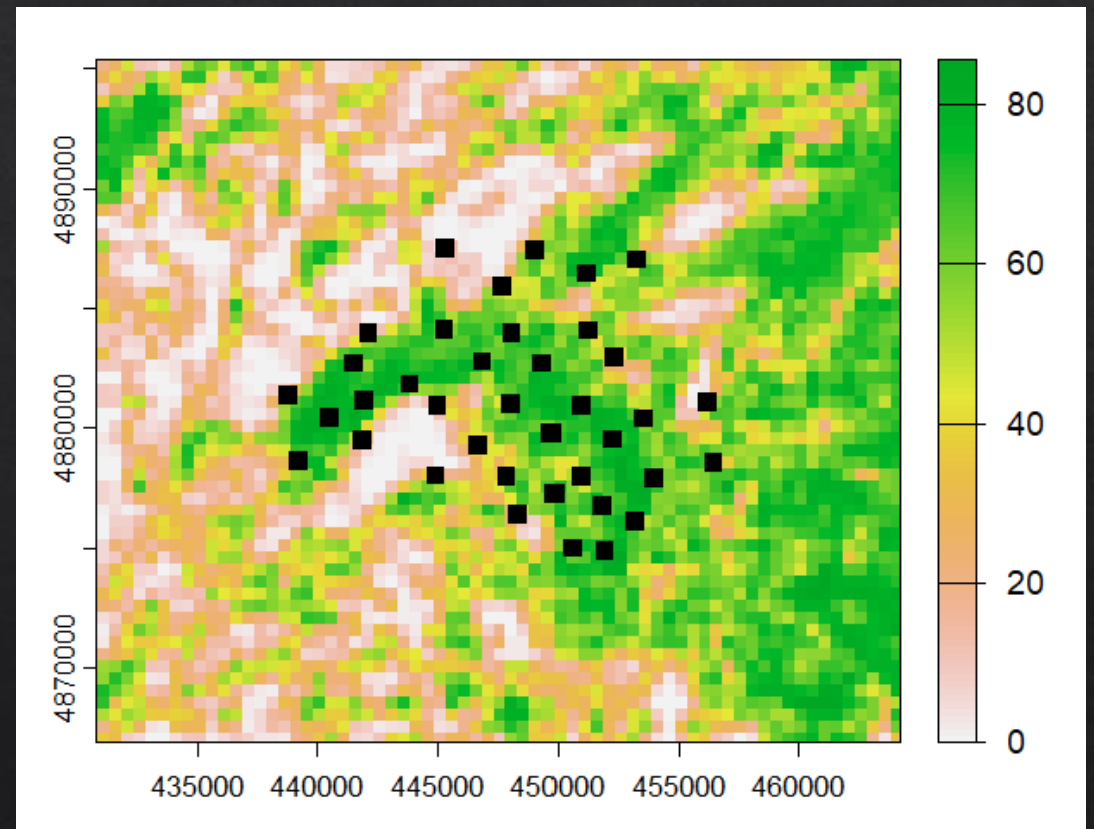
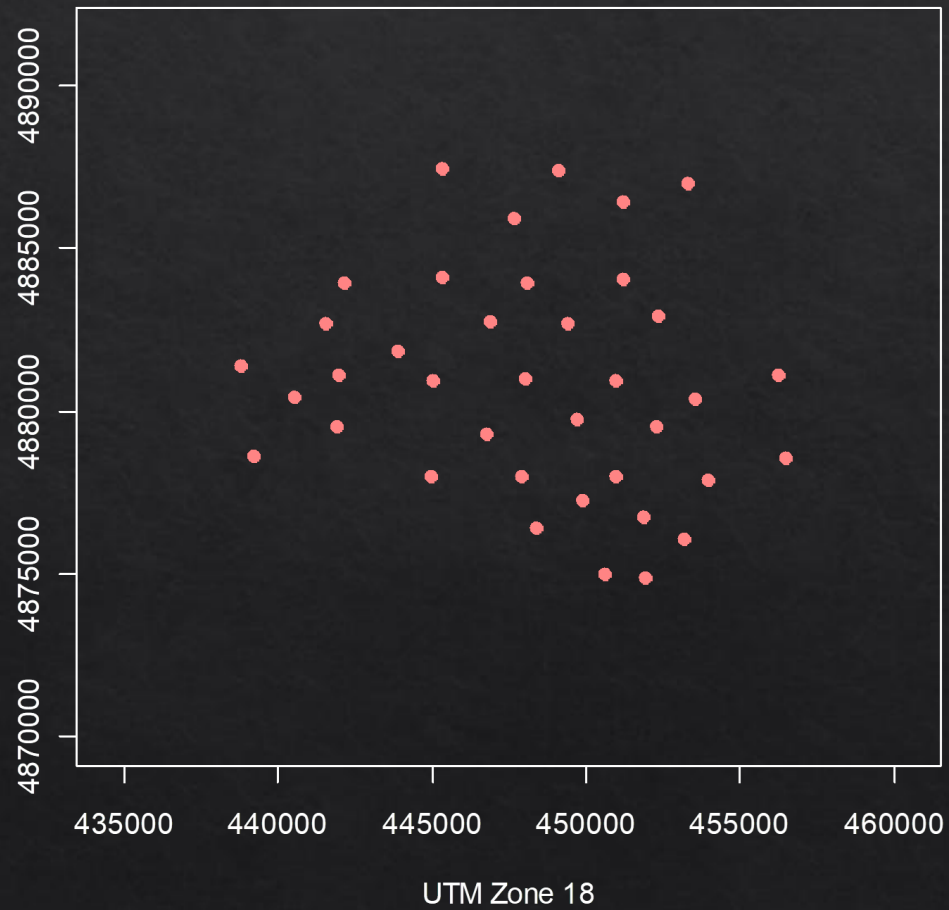
- ◆ Set 38 baited traps
- ◆ Checked weekly for 8 weeks
- ◆ 47 unique individuals identified
- ◆ Use a 3-D encounter history



The data files

- ◆ `load(beardata.rda)`
- ◆ `beardata`
 - ◆ **trapmat** - trap id, x location, y location (in UTM's in kms)
 - ◆ **bearArray** - 3D array of captures by individual, trap, occasion
 - ◆ **flat** - encounter data formatted for the R package secr
 - ◆ **sex** – the biological sex assigned to individuals
- ◆ `forest_cover.tif`

Data files



Bear Models

- ◆ Behavioral response

- ◆ Might expect bears to respond to the baited traps (e.g., bears like donuts, they will go back for more)

- ◆ Movement

- ◆ Expect that males and females have different home range sizes

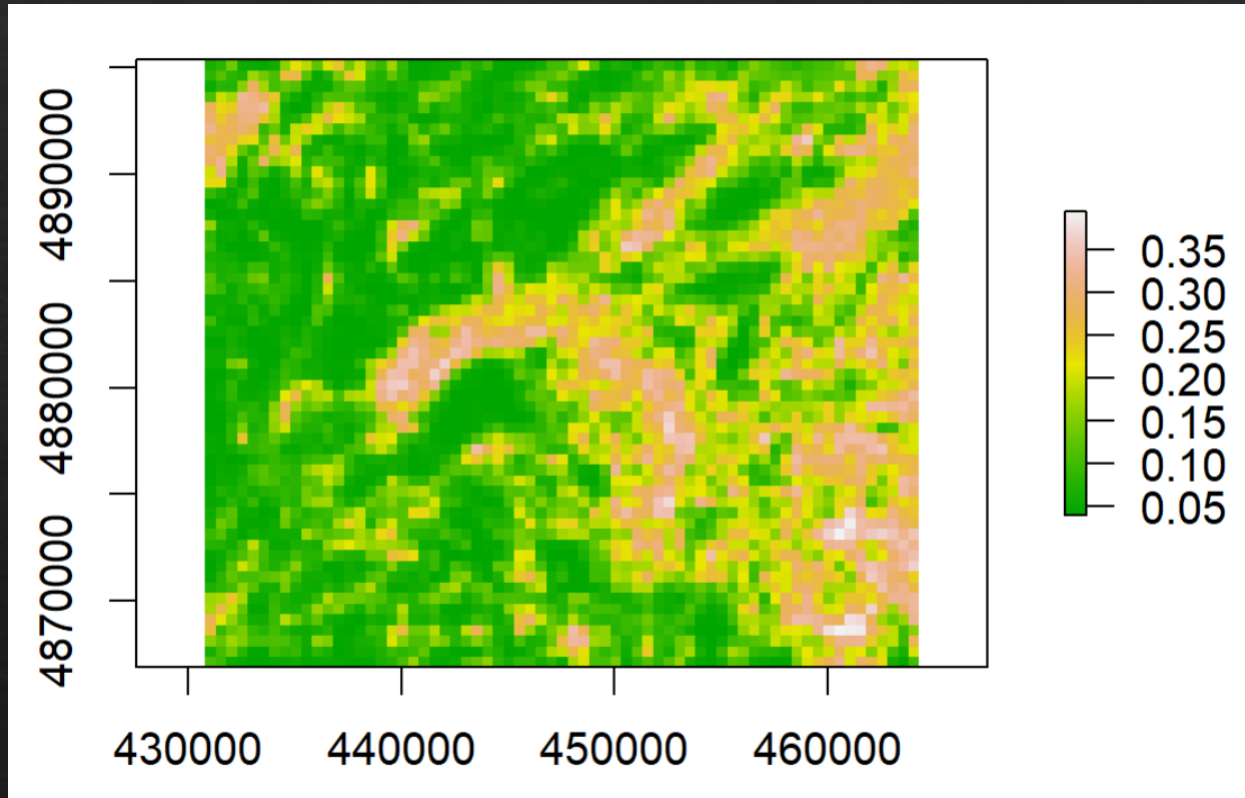
- ◆ Density

- ◆ Expect that bears will prefer more forested areas

Bear Models

- ◆ ModelSCR_0.R - no covariates
- ◆ ModelSCR_behavior.R – behavioral response
- ◆ ModelSCR_sex.R – detection parameters vary by sex
- ◆ ModelSCR_habitat.R – density varies by forest cover

Bear density



- ◇ SCR habitat file allows you to estimate density as function of a spatial covariate.
- ◇ This file also includes
 - ◇ User defined function for NIMBLE
 - ◇ Alternative samplers to improve mixing

Steps

- ◇ Select the correct observation model for your data
 - ◇ Make sure your data are formatted for this structure
- ◇ Can include individual, trap, and occasion specific covariates (baited traps, on-off roads, seasonal effects on detection, age and sex classes of individuals, etc.)
 - ◇ Adjust either g_0/λ_0 or σ accordingly
- ◇ Carefully choose the state space and data augmentation

```

modelSCR0<- nimbleCode( {

  #priors
psi~dunif(0,1) #inclusion probability
g0~dbeta(1,1)  #detection probability
sigma~dunif(0, 6) #scale parameter, units of kms
sigma2<-sigma*sigma

for(i in 1:M){ #M observed + augmented individuals
  z[i] ~ dbern(psi)      #indicator if individual is in the population
  s[i,1]~dunif(Xl,Xu)    #activity centers X coord
  s[i,2]~dunif(Yl,Yu)    #activity centers Y coord

  for(j in 1:J){ #J traps
    d2[i,j]<- pow(s[i,1]-X[j,1],2) + pow(s[i,2]-X[j,2],2) #calculate the distance
    g[i,j]<- z[i]*g0*exp(-d2[i,j]/(2*sigma2))             #detection function
    y[i,j] ~ dbin(g[i,j], K)
  }
}

N<-sum(z[]) #number of individuals in the state space
D<-N/area   #density of individuals in the state space
})

```

Observation process

```
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})

```

Choosing the state space?

- ◆ Must be large enough that any animal who could have encountered the trap array is included
- ◆ A little bit trial and error
 - ◆ Should probably be about 3 times σ (sometimes more)
 - ◆ If you buffer too much, density will remain constant, but you will increase computation time
 - ◆ If you buffer too little, the density estimate will be influenced
- ◆ Can add habitat masks and other state space configurations

Data augmentation

- ◆ Must provide a large enough value to not influence the estimate of N
- ◆ Check the posterior distribution of N or ψ to ensure it is not truncated
- ◆ The data augmentation is related to the buffer: if you increase the state space, you'll need to increase the number of augmented individuals

Move to R

Summary

- ◆ Design a study considering the trap spacing and movement of your species
- ◆ Select the correct observation model for your data
 - ◆ How many times can an individual be captured in a trap during any given sampling occasion?
- ◆ Can include individual, trap, and occasion specific covariates (baited traps, on-off roads, seasonal effects on detection, age and sex classes of individuals, etc.)
- ◆ Carefully choose the state space and data augmentation

Acknowledgements

- ESA Statistical Ecology Section
- Jody Peters
- Grace DiRenzo
- Tom Qui and Jeff Doser

Institutions

- University of Washington
- North Carolina State University
- USGS
- Patuxent Wildlife Research Center

- J. Andrew Royle
- Rahel Sollmann
- Nathan Hostetter
- Richard B. Chandler
- Sarah Bassing



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